ModelIt!

Student Activity Pack

How Do Animals Use Their Body Parts?

The Science of Structure and Function

Grade 1 | NGSS 1-LS1-1 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw an animal and circle one body part. Write or tell what job that body part does.

2 Circle the body part that helps a bird eat: wings, beak, tail feathers.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Body Part Size | Ability to Get Food

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Long Beak

Body Part Size set to long beak

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Short Beak

Body Part Size set to short beak

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do animals use their body parts to help them survive? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design an Animal Tool

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YOUR ENGINEERING MISSION

Build a tool inspired by an animal body part that helps solve a problem, like picking up small objects.

THE SCENARIO

Scientists need your help! They want to pick up small beads from a cup without using their fingers. Look at animal body parts for ideas. Can you design a tool that works like an animal's beak, claw, or trunk to grab the beads?

Engineering Design Thinking

1. WHAT ANIMAL BODY PART INSPIRED YOUR TOOL DESIGN?
2. WHAT MATERIALS WILL YOU USE TO BUILD YOUR TOOL?
3. HOW WILL YOU TEST IF YOUR TOOL CAN PICK UP SMALL BEADS?
4. CAN YOU MAKE YOUR TOOL WORK BETTER BY CHANGING ITS SHAPE OR SIZE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomedical Engineers design tools and devices inspired by animal body parts, like robotic arms that work like elephant trunks. They earn $50,000-$95,000/year.